**Section 1 – Sprint Goal**

Write ONE clear, demonstrable Sprint Goal.

It must:

- Be specific
- Be measurable
- Be demonstrable
- Add value toward the Product Goal

Sprint Goal:

Create the basic skeleton for the app, integrating the device's camera and the Open Food Facts API so that we can successfully scan a product barcode and retrieve its name and primary nutritional data.

How will you demonstrate this goal at Sprint Review:

We will open the app on an Android emulator or device, point the camera at a food product's barcode, and show a basic UI screen that displays the product's name and its raw ingredient list pulled directly from the database.

**Section 2 – Selected Product Backlog Items**

List the backlog items select for Sprint 1.

These must:

- Be small enough to complete in one Sprint
- Be demonstrable
- Move the product toward MVP
- For each item specify if it's size – S/M/L

Item 1: Open the app and access the camera scanner so the user can prepare to scan a product (S)

Item 2: Scan a barcode and pass the decoded string to the Open Food Facts API (M)

Item 3: Have the API return and display basic product information (Name, Brand, Calories) (M)

Item 4: Set up basic Allergen Profile UI (common allergens like Peanuts/Dairy) and a basic UI screen displaying all the results (M)

(List additional items if realistic)

Why is this amount of work realistic?

Since we have Java experience, the API calls shouldn't be too difficult. Since we are also just focusing on the barcode rather than complex OCR, from a technical standpoint, this will be a lot easier. We are just focusing on the core features for now which will provide a skeleton for all of the future AI features.

**Section 3 – Task Breakdown**

Break each backlog item into tasks.

Tasks must:

- Be 1 day or less
- Be assignable
- Be concrete

Backlog Item 1 Tasks:

- Initialize Flutter/Kotlin project and configure Camera permissions
- Add a “scan” button on the home screen that opens the camera
- Create the UI for the camera

Backlog Item 2 Tasks:

- Add a barcode scanning library like google ml kit
- Store the barcode string in a variable for the API request

Backlog Item 3 Tasks:

- Make the http get request for the Open Food Facts API with the scanned barcode
- Parse JSON response to extract relevant information like product name and nutrients

Backlog Item 4 Tasks:

- Create a simple list of checkboxes (Peanuts, Dairy, Gluten, etc.) where users can toggle their allergies
- Make sure allergens saved even after app closes
- Create a simple layout that displays all of the pertinent information
- Compare ingredient list to allergen profile and trigger warnings if a match is found

Repeat for each Backlog Item

**Section 4 – Ownership and Accountability**

Each task must have a primary owner:

Owner assignments:

| Task | Owner |
|---|---|
| Camera UI and Permissions | Arda |
| Barcode Library | Leo |
| API Request and JSON Parsing | Siddharth |
| Results Screen and Navigation | Varun |

All members must have development tasks.

Who will be your Scrum Master for this Sprint?

Varun Hari

**Section 5 – Definition of Done Confirmation**

Confirm:

- Does every selected item meet Definition of Done before more to Done?
- Is peer review required?
- Is demonstration required?

Team agreement:

We agree that no item will be considered complete unless it meets the Definition of Done.

Signatures:

Arda

Siddharth

Varun

Leo

**Section 6 – Risk Check**

What could cause this Sprint to fail?

- Risk 1: The school Wi-Fi might block the Open Food Facts API or the emulator might not connect
- Mitigation: Test the API endpoint on the school network early on and make backup plans as necessary

- Risk 2: Testing a camera on a laptop emulator can be buggy
- Mitigation: Deploy to a physical Android device for testing as soon as possible